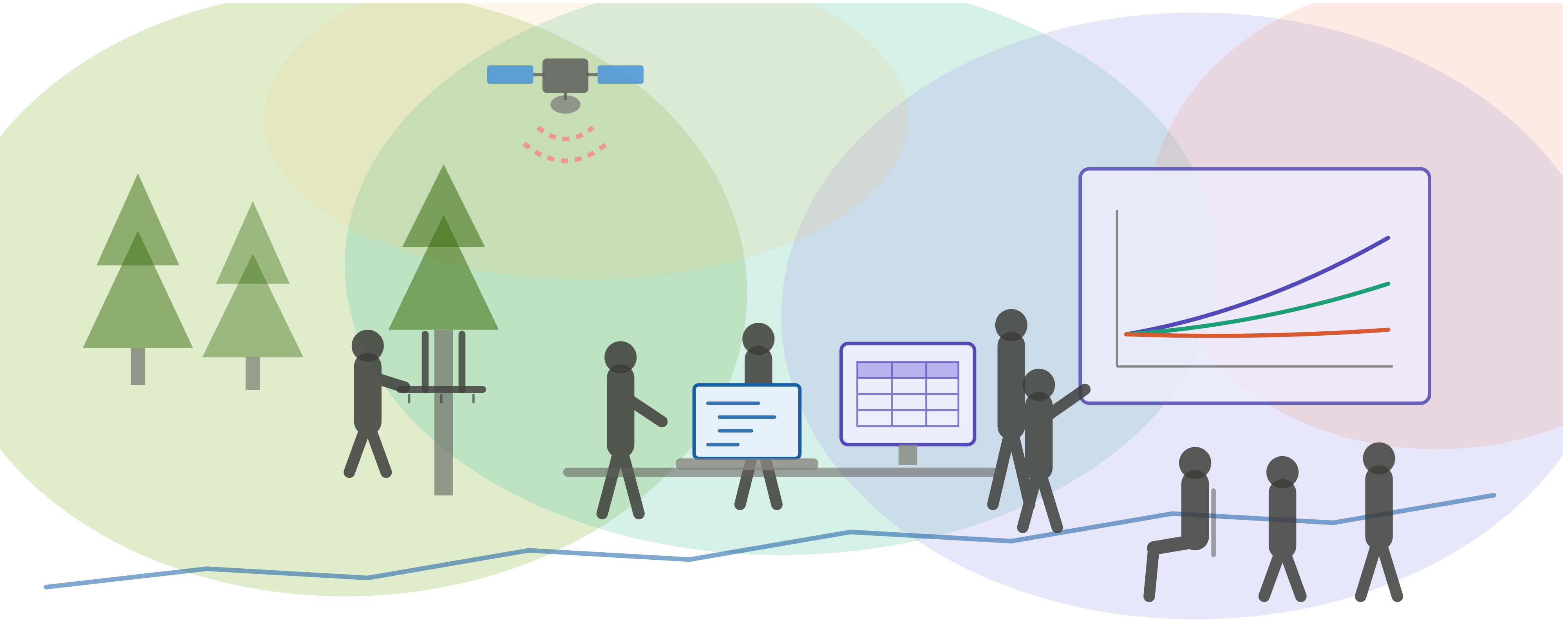
AgroParisTech, Forest Information and Resource Strategies (FRS) Mastèr
Model-assisted prospective for forests and forest sectors **module**

Part 1: Modelling Foundations



Antonello Lobianco
Bureau d'Economie Théorique et Appliquée (BETA)
AgroParisTech

"All models are wrong, but some are useful"
– George Box (Box 1976) (British statistician)

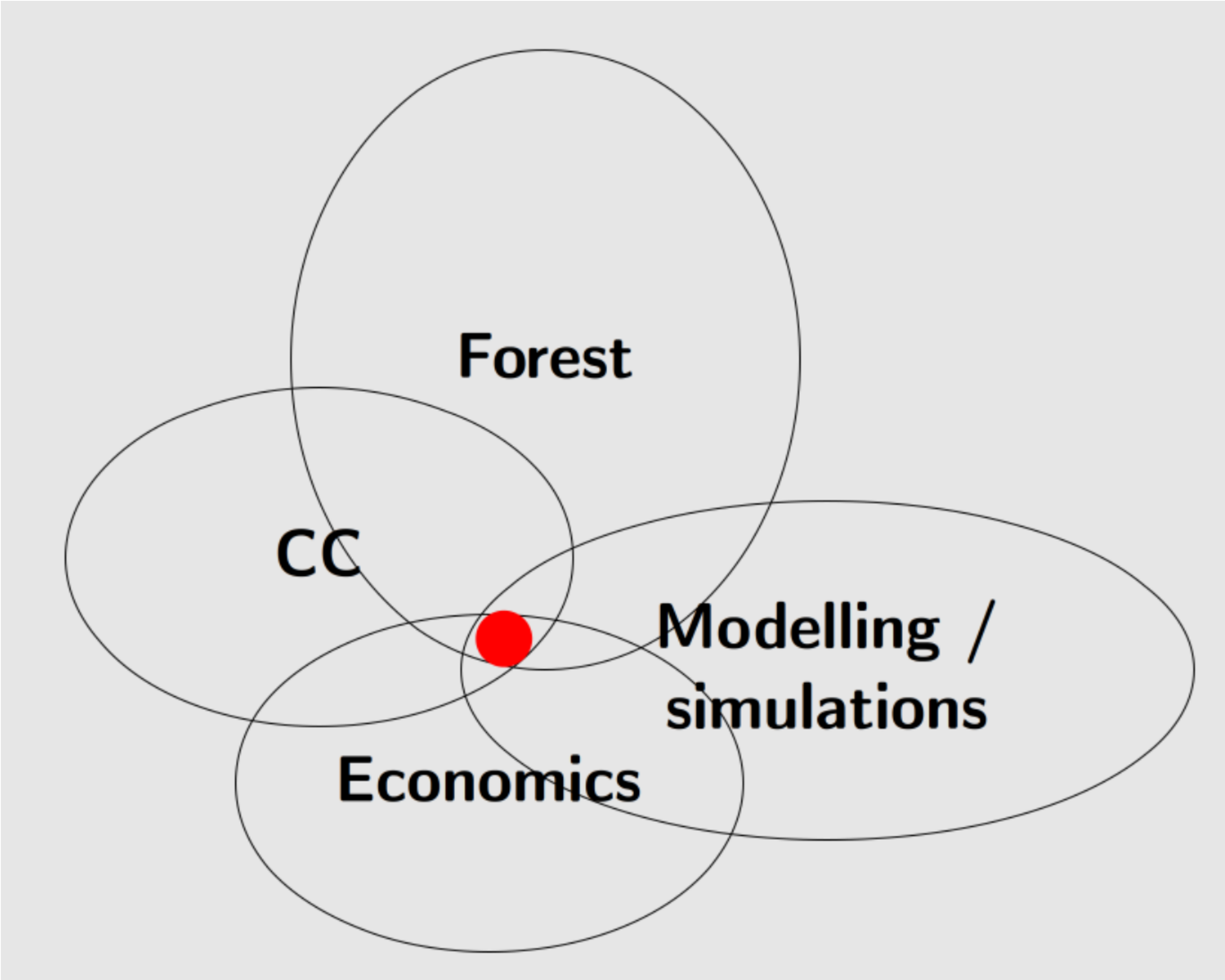
About myself

- **Forest and natural resource economist** @ Bureau d'Economie Théorique et Appliquée (BETA) Nancy
- Master in Forest and Environmental Sciences and PhD in agricultural policies
- My work involves a lot of **modelling and simulations**
  - agent-based farm-level to regional simulations → RegMAS (Regional Multi Agent Simulator)
  - bio-economic models of the forest sector (forest dynamics, timber markets equilibrium, carbon cycle) → FFSM (French Forest Sector Model)
  - more recently: machine learning algorithms, https://github.com/sylvaticus/betaml.jl

Research projects I am currently involved in:

- PEPR *Carbone et Écosystèmes continentaux* (FairCarbon):
  - SLAM-B (*Scénario labs et évaluation intégrée par modélisation pour le développement de la bioéconomie*): Gestion des forêts et nouveaux usages du bois en France
- PEPR FORESTT:
  - REGE-ADAPT: cc → regeneration
  - X-RISKS: CC → multirisk

About myself



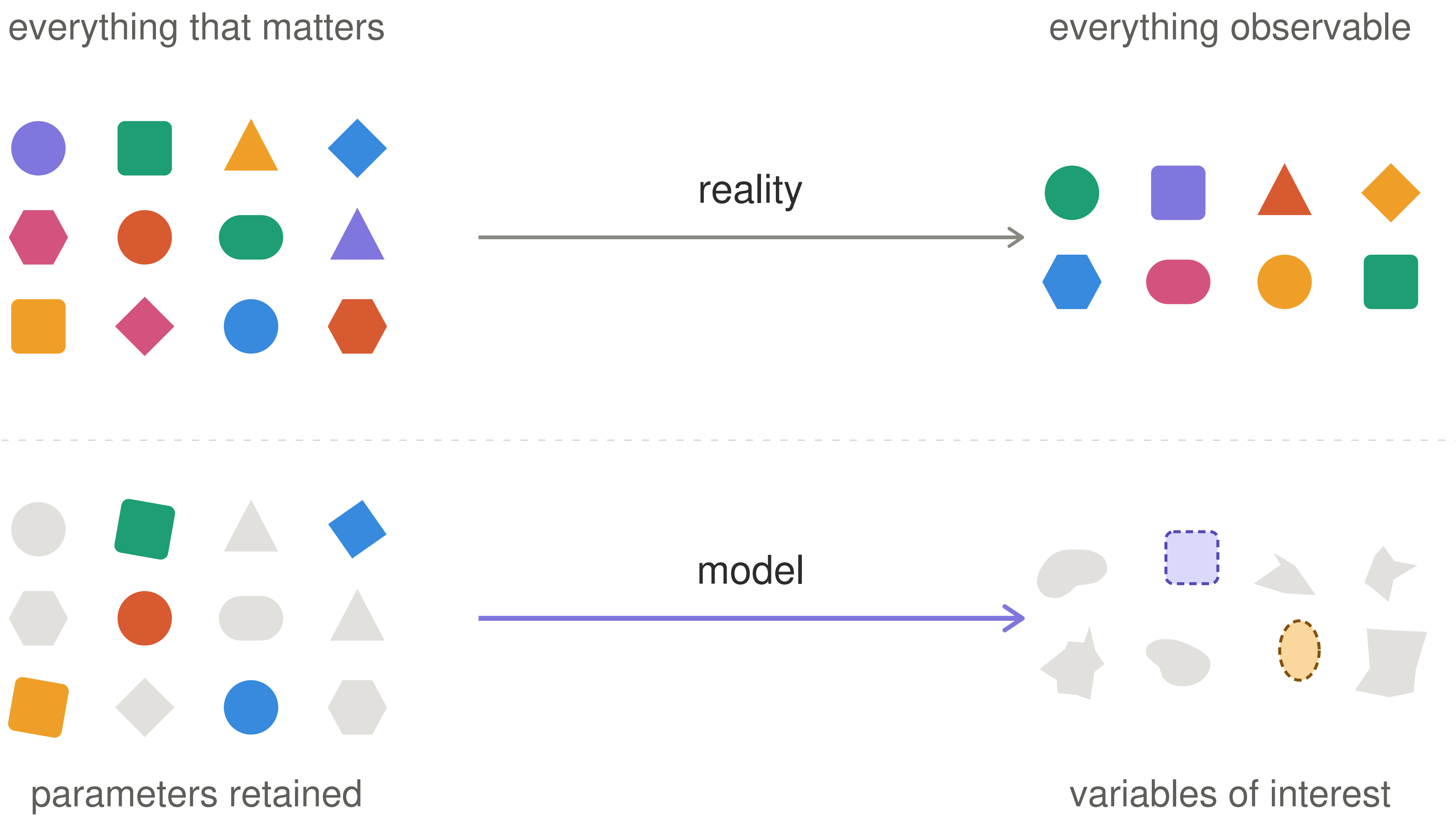
Course program

1. **Modelling foundations** (today; 3h): main concepts and principles of modelling, model typology, scenario building, model evaluation and validation, uncertainty and sensitivity analysis, integration and coupling of models, reproducibility
2. **Programming foundations** (7 Sept; 6h): introduction to programming concepts and techniques for model development and implementation, introduction to VSCode, Python and R
3. **Advanced programming** (20, 21, 22 and 23 Oct AM; 12h): advanced programming concepts and techniques for model development and implementation, including good practices and development workflows (environment segregation, code versioning, testing, documentation/literate programming, packaging, AI & coding). Based on Julia
4. **Group work** (20 and 22 Oct, PM; 6h++): practical exercises and group work on model development and implementation
5. **Case studies** (21 Oct AM, 16-18 Dec, 22 Janv, 26 Janv PM; 15h): analysis of real-world applications of model-assisted prospective studies in the forest sector: IGN MARGOT, FFSM, EFISCEN, etc.

Part A

Model definition

What is a Model?



Why do we model?

A model is not built to be *true*; it is built to be **useful for a purpose** (Box 1976). Naming that purpose is the first modelling decision, and the one most often skipped (Edmonds et al. 2019).

- **Predict** a quantity we will *later* observe, or a quantity we cannot observe directly or that is expensive to observe
- **Understand or explain**: *if* some assumptions hold, *then* something follows; which processes are sufficient to produce the pattern we observe
- **Explore** the consequences of assumptions, including implausible ones
- **Support a decision**: rank options, expose trade-offs
- **Communicate** and structure a debate between stakeholders
- **Train** people, or **describe** a system compactly
- **Discipline our own thinking**: writing the model reveals what we did not know we assumed

Warning

Be careful not to sell a model as a crystal ball: a model built to *explore* a mechanism being quoted as if it had *predicted* a number (**purpose mismatch**)

From Reality to Code

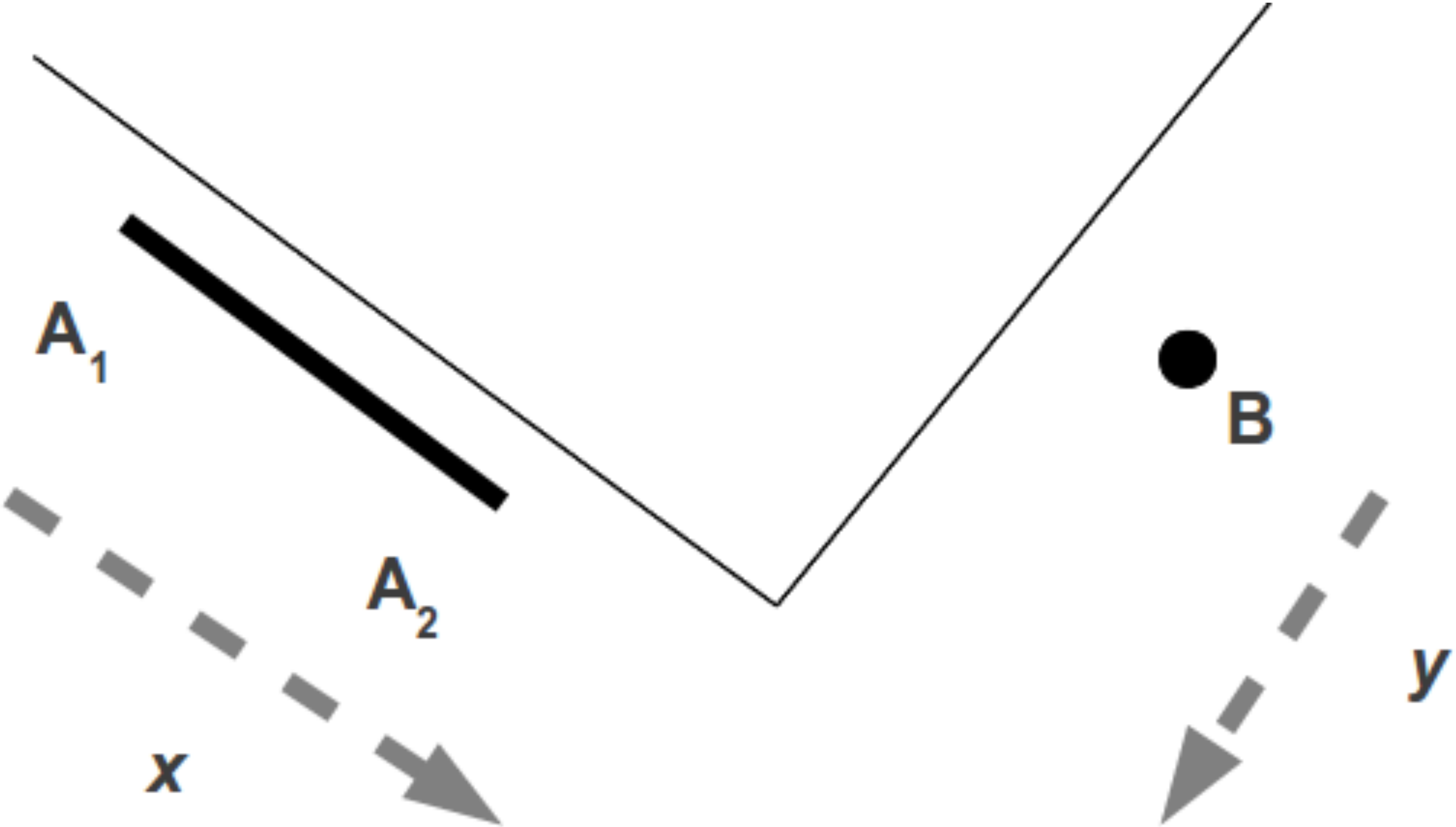
- reality

Reality → model → math. form. → code

From Reality to Code

- model

Reality → model → math. form. → code



A model is useful if it helps us to answer a research question, in this case whether a crash will happen.

From Reality to Code

- math

Reality $\rightarrow$ model $\rightarrow$ math. form. $\rightarrow$ code

We need to define which elements to retain from our observation of the reality (inductive reasoning), which assumptions to make and which data we need, and which reasoning to apply (deductive part) in order to answer the research question.

Info needed:

$x_{A1}, x_{A2}, y_B, v_A, v_B$

Assumptions:

v_A, v_B constant

Model:

physical space/time/speed relation

$t_{A1} = \frac{x_{A1}}{v_A}$ $t_{A2} = \frac{x_{A2}}{v_A}$ $t_B = \frac{y_B}{v_B}$ $\Rightarrow$ crash iff $t_{A2} \leq t_B \leq t_{A1}$

From Reality to Code

- code

Reality → model → Math. form. → code

Example of an implementation in Julia

- **Download and doc**: https://julialang.org
- **Tutorial**: https://www.gitbook.com/book/sylvaticus/julia-language-a-concise-tutorial/details

```julia
function determineClash(xa1, xa2, yb, va, vb)
  ta1 = xa1/va
  ta2 = xa2/va
  tb = yb/vb
  if (ta2 <= tb <= ta1)
     return true
  else
     return false
  end
end
```

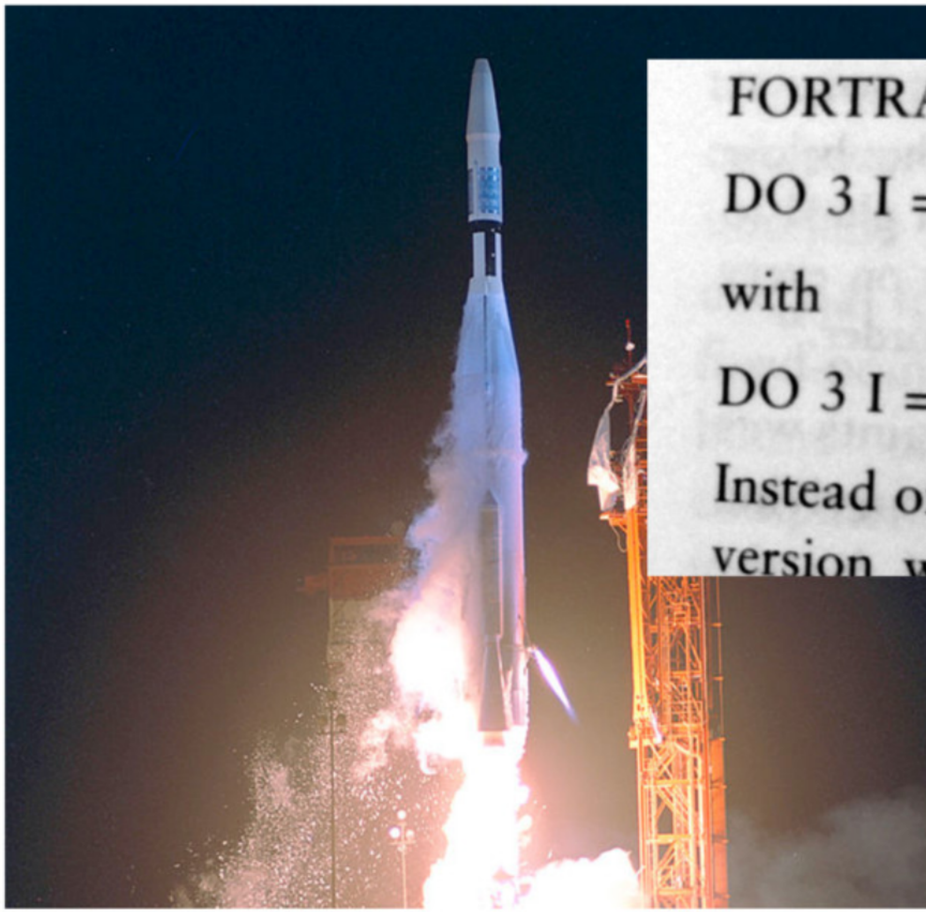
```julia
determineClash(8,5,12,3,6) |> println # true
determineClash(8,5,12,3,4) |> println # false
```

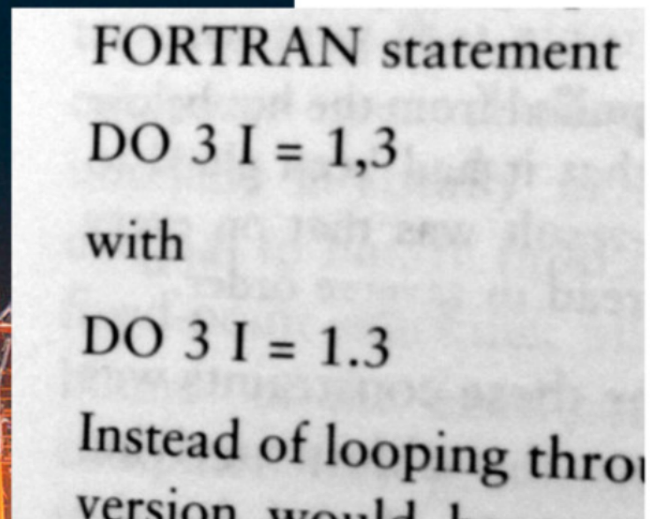
Models can be "wrong" either in the translation of elements that matter in the modelling idea (assumptions) or in their formalisations (e.g. Mariner 1, 1962 satellite, Mars Climate Orbiter 1998)

Famous errors

- syntax

Programming is error-prone





FORTRAN statement

DO 3 I = 1,3

with

DO 3 I = 1.3

Instead of looping thro

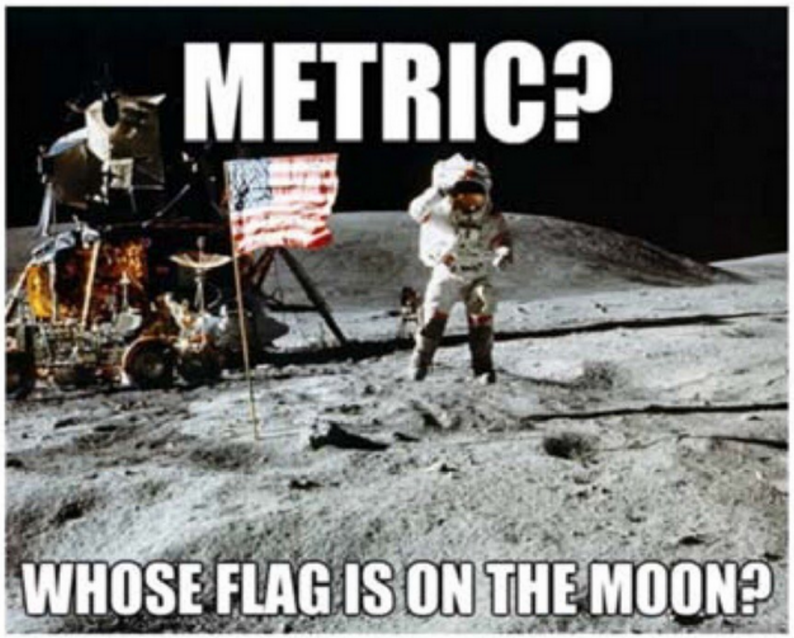
Mariner 1
1962

Famous errors

- unit system

F...A...I...L...U...R...E... (Famous bugs)

- Mars Climate Orbiter (1998)
  - $655M space probe crashes due to wrong units system.



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Slide 3

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Part B

Model typologies

What kind of model?

There is **no official taxonomy of models**. What exists is a small set of design choices; a "model type" is just a recurrent combination of them (Kelly (Letcher) et al. 2013).

Dimension		**Range of choices**
Purpose		explain / predict / explore / decide / communicate / train
Mechanism	✓	correlative (data-driven) ↔ process-based (causal)
Aggregation		representative agent ↔ heterogeneous individuals
Space		none / regions / grid / individual location
Time		static / comparative static / dynamic (recursive / myopic or intertemporal / perfect foresight)
Behaviour		none / optimising / adaptive / rule-based
Stochasticity	✓	deterministic ↔ stochastic
Data intensity		low (parameters from literature) ↔ high
Code implementation	✔	Single equation, Compiled language, Interpreted language
Availability		Not available, Under request, Open source, Free software
Reproducibility	✓	non-reproducible, reproducible with instructions / automatic / including input data

Warning

Often the evil is in the details!

Statistical models

Fit a **functional form** to observed data and use the estimated parameters as the object of interest.

$$y_i = \beta_0 + \beta_1 x_{1i} + \ldots + \beta_k x_{ki} + \varepsilon_i$$

What they are good at

- quantifying an effect, with a **confidence interval** around it (elasticities, growth increments, dose-response)
- testing a hypothesis derived from theory
- explaining causal relationships (tricky!!!)

What they are not good at

- extrapolating outside the observed domain
- answering "what if the rules of the game change?" - the estimated parameters embed the policy regime in force when the data were generated (Lucas 1976)

Statistical models

- explain or predict?

Breiman (2001) describes two cultures:

Data modelling

- assume a stochastic data-generating process
- estimate its parameters
- validate with goodness-of-fit tests
- the *model* is the answer

Algorithmic modelling

- treat the data mechanism as unknown
- optimise predictive accuracy
- validate out-of-sample
- the *prediction* is the answer

- A model with high **explanatory** power can have poor **predictive** power (and the reverse).
- Collinear regressors ruin inference on β but may not hurt prediction at all.
- R^2 on the estimation sample answers neither question well.
- the distinction between *decide* and *predict* is often blurred in policy applications

Predictive models (non-parametric / machine learning)

We do **not** need to hardcode the rules of the system; the algorithms *learn* them from the data (Lobianco 2025).

- **Supervised**: predict a *label* `y` from *features* `x`
  - `y` continuous → regression
  - `y` categorical → classification
- **Unsupervised**: find structure without labels
  - clustering, dimensionality reduction
- **Reinforcement**: find the optimal sequence of actions towards a goal, with feedback only on the final state

A surprisingly small number of general-purpose algorithms fit a very wide range of problems:

- tree-based algorithms: decision trees, random forests
- neural networks
- k-means, GMM, PCA, autoencoders

The division is nowadays didactic: modern systems (e.g. large language models) mix all three.

Predictive models

- parameters vs hyperparameters

- **Parameters** are learned from the training data.
- **Hyperparameters** are fixed *before* training but still change performance: number of neurons per layer, number of trees in a forest, how *long* to train.

The objective is **not** to learn the relation between the given `x` and the given `y`. It is to learn the *general* relation in the population from which they were sampled.

Overfitting (too much *variance*)

Too many neurons, too much training: we learn the noise of *this* sample. Different training data → very different parameters.

Underfitting (too much *bias*)

Model too simple, or trained too little: the relation is not learned at all.

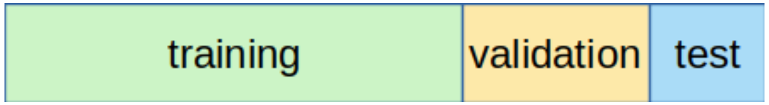
Techniques that improve generalisation at the cost of training performance are called **regularisation**.

Predictive models

- how do we choose the hyperparameters? 3-ways split

Split the data **three** ways:

- **training set** - learn the parameters, given a set of hyperparameters
- **validation set** - out-of-sample score, used to *choose* the hyperparameters
- **test set** - final, honest judgement, used **once**

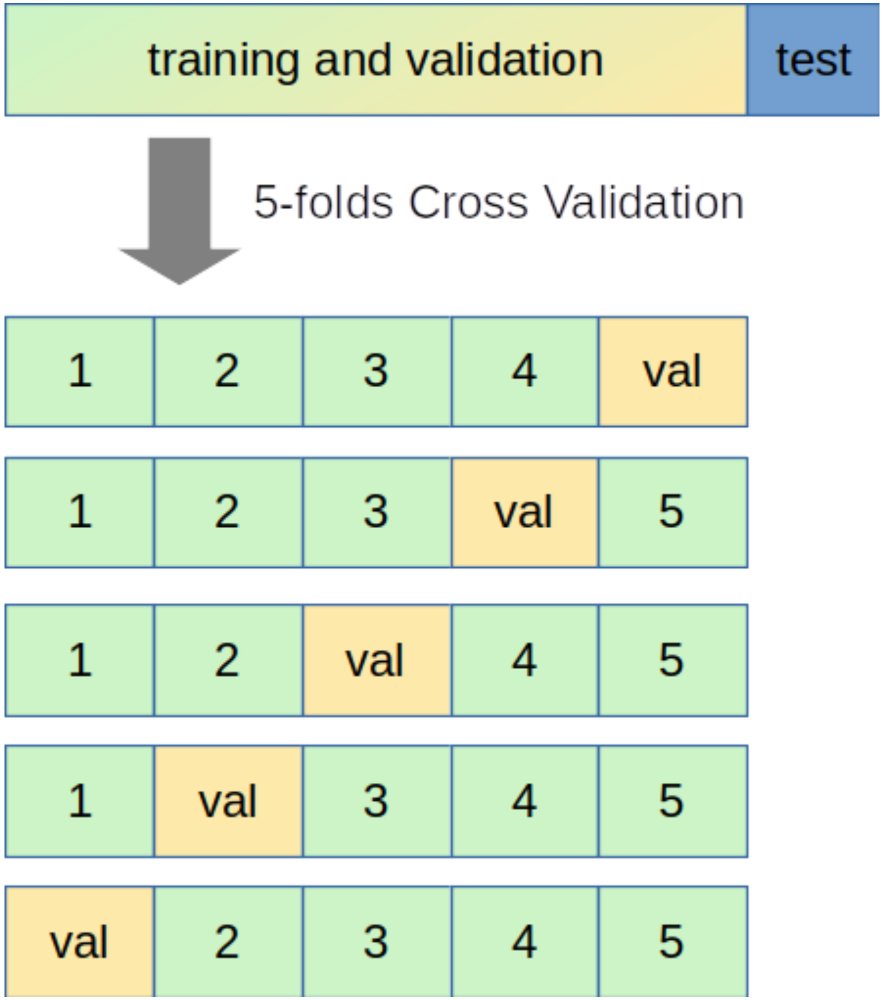


You cannot judge the model on the validation set: the hyperparameters were fitted on it.

Predictive models

- how do we choose the hyperparameters? Cross-validation

K-fold cross-validation: partition the train/validation pool into K subsets; train on $K-1$, score on the remaining one; repeat K times and average.



Useful whenever the algorithm has a stochastic component (initialisation, sampling) or the data is limited.

Predictive models

- cross-validation example

```
# BetaML: cross-validation over a user-defined block
x = rand(500,2); y = 2*x[:,1] + 3*x[:,2] + randn(500)
sampler = KFold(nsplits=5)
for md in [1, 2, 4, 8, 16, 32, 64]
    (μ, σ)  = cross_validation([x, y], sampler) do trainData, valData, rng
        (xtrain, ytrain), (xval, yval) = trainData, valData
        m  = DecisionTreeEstimator(max_depth=md)
        fit!(m, xtrain, ytrain)
        relative_mean_error(yval, predict(m, xval))
    end
    println("max_depth=$md: mean error=$μ ± $σ")
end
```

```
max_depth=1: mean error=0.4590803030765838 ± 0.016850740218957504
max_depth=2: mean error=0.42359351497954006 ± 0.035925914073438134
max_depth=4: mean error=0.35264984209027245 ± 0.037169074993346008
max_depth=8: mean error=0.41593637465450894 ± 0.02049019967776637
max_depth=16: mean error=0.4511119184675425 ± 0.04515625903196677
max_depth=32: mean error=0.4305527994177497 ± 0.02603276496623473
max_depth=64: mean error=0.4610102521440719 ± 0.02680524249457227
```

Predictive models

- strengths and limits for prospective work

Strengths

- non-linearities and interactions learned from data, not imposed
- high-dimensional inputs (remote sensing, inventory plots)
- no functional form to defend
- variable correlation is not an issue
- feature importance gives a ranking of drivers (`FeatureRanker`: mean decrease in accuracy, Sobol' contributions)

Limits

- interpolation machines: the training domain *is* the domain of validity
- no causal content ⇒ **cannot** answer counterfactual policy questions
- "black boxes": for high-stakes decisions, prefer models that are interpretable by construction rather than explained after the fact (Rudin 2019) (biased data -> biased outcomes)
- hungry for data

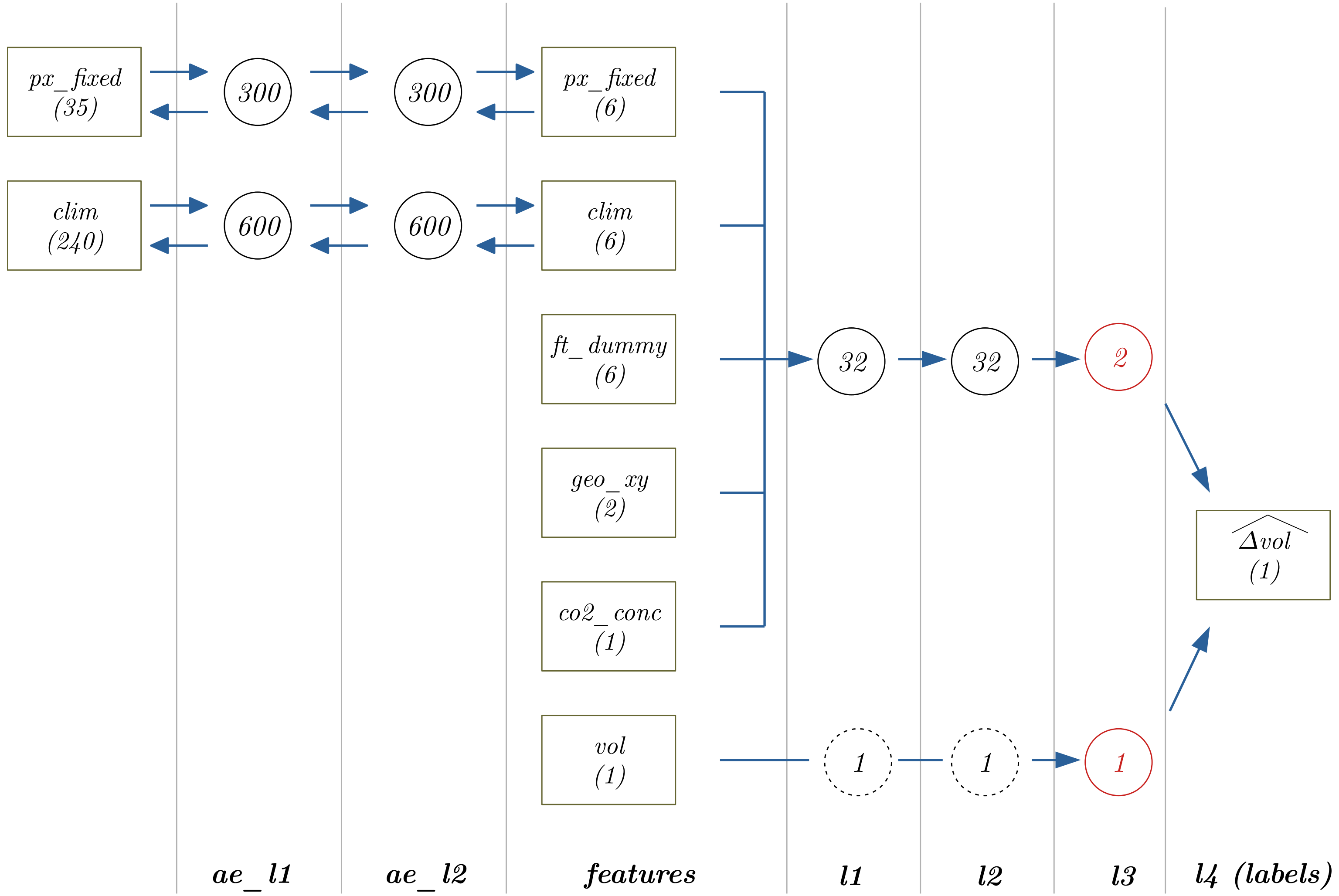
Hybrid modelling is the promising middle ground: use ML for the components we cannot write down (e.g. a mortality response), keep the process-based skeleton for consistency with physical and biological constraints (Reichstein et al. 2019).

Hybrid model

- Verhulst model of stand growth

$$\frac{\delta V}{\delta t} = rV - \frac{r}{K}V^2$$

Growth model architecture



Mechanistic / process-based models

Represent the **causal mechanisms** of the system, with parameters that have a physical or biological meaning.

$$\mathrm{NPP} = \varepsilon \cdot \varphi_{p.a.} \cdot f(\mathrm{VPD}, \theta_{soil}, \mathrm{age}) \qquad \text{(3-PG, simplified)}$$

- Can, in principle, be run **outside** the observed domain: that is the whole point
- Parameters transfer across sites and species (with caution)

Examples: 3-PG (Landsberg and Waring 1997) (stand level forest growth), JABOWA (Botkin et al. 1972; Bugmann 2001) (gap model), LANDIS-II (Scheller et al. 2007) (forest landscape), ORCHIDEE (Krinner et al. 2005; Naudts et al. 2015) (global vegetation), STICS (Brisson et al. 1998) (crop growth), AMG (Clivot et al. 2019) (soil carbon)

Mechanistic models

- how much complexity?

Bugmann and Seidl (2022) reviewed 28 models of long-term forest dynamics on 52 attributes:

- five distinct clusters, from very simple to complex on every axis
- most models in use today are **not balanced**: some processes are rendered in great detail while others in the same model stay crude
- this reflects in part model purpose, but also code legacy, the developers' own domain/preferences and the "prevailing spirit of the epoch"

Their conclusion: model diversity is *useful*. Agreement between structurally **different** models is evidence of robustness; agreement between **similar** models can convey a false sense of certainty.

Economic equilibrium models

- Model the aggregated supply and demand per product and region and their interaction (markets)
- Prices are **endogenous**, determined by the market clearing condition, together with demand, supply, consumption, imports and exports
- Solved by matching supply/demand equilibrium prices or quantities or often as a welfare-maximisation problem
- Typically divided into two categories:
  - **Partial equilibrium models**: take into consideration only a part of the market (a single product, or a sector), ceteris paribus, to attain equilibrium. They assume that, on the one hand, the price of every other good or the wealth one has does not change (and hence is given), and on the other hand that what happens on the market one wants to analyse has no effect on other markets (simpler and more "depth" on the specific market but potentially unrealistic)
  - **General equilibrium models**: the analysis includes the entire economy. Every market has an effect on every other market and therefore one has to model every market simultaneously (realistic but harder to solve)
- Can be coupled with biophysical models to form Integrated Assessment Models

Economic equilibrium models

- caveats and examples

Caveats:

- Strong assumptions on the institutional framework (e.g. competitive markets) and agents' behaviour (e.g. rational)
- Only the market equilibria are modelled, not the way to reach them.
- Results strongly depend on the parameters (elasticities) and chosen functional forms of supply/demand

Used for: carbon taxes (DICE/RICE (Nordhaus 1992)), fuelwood subsidies (FFSM (Caurla et al. 2013)), timber trade (GFPM (Buongiorno 2015), GFPMX (Buongiorno 2021))

Simulation models

When the system cannot be solved **analytically**, and statistical analysis is defeated by its chaotic behaviour, there is a third way: state the assumptions, run the system, and *observe* what emerges.

Complicated $\neq$ complex

- *complicated*: Latin root **plic**, "to fold", "to hide" - the opposite of *simple*
- *complex*: Latin root **plex**, "to weave" - the opposite of *independent*

A complex system has many interwoven, interdependent components, producing a whole that differs from the sum of its parts.

Consequences

- non-linear and highly dynamic
- high sensitivity to initial conditions
- **emergent properties**, not deducible from observing single components
- deterministic, yet apparently driven by random forces

Economic systems add one further difficulty over physical and biological ones: individual **strategic behaviour** (Arthur 1999).

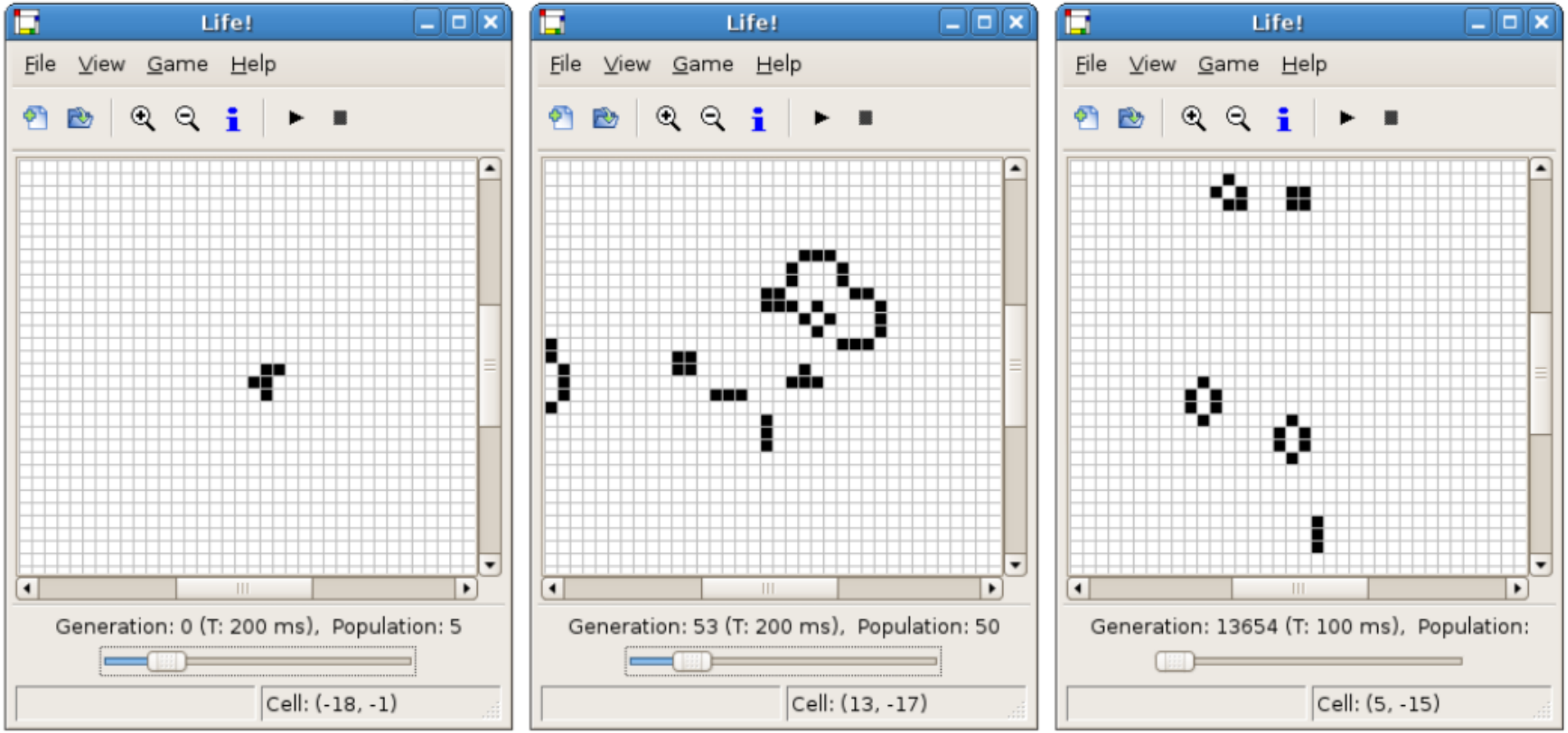
Simulation models

- cellular automata

Discrete, spatially extended dynamic systems: adjacent cells on a grid, each with an internal state from a finite set, updated by a **transition function** of the neighbours' states. Every cell obeys the *same* rule; updating is usually simultaneous. Invented by von Neumann in the early 1950s to study self-replication (Neumann 1966).

Applications: wildfire propagation (Karafyllidis and Thanailakis 1997), lava flows, cancer-cell spread, land-use change, farm structural change (Balmann 1997)

Well-known example: **The Game of Life** (Gardner 1970)


Life!
File View Game Help
Generation: 0 (T: 200 ms), Population: 5
Cell: (-18, -1)
Life!
File View Game Help
Generation: 53 (T: 200 ms), Population: 50
Cell: (13, -17)
Life!
File View Game Help
Generation: 13654 (T: 100 ms), Population:
Cell: (5, -15)

Simulation models

- Example 1: Game Of Life

Rules

- *Space*: infinite or toroidal grid.
- *Time*: discrete, synchronous updates.
- Two *states*, live or dead:

1. a live cell with ≤ 1 live neighbour dies (loneliness)
2. a live cell with ≥ 4 live neighbours dies (overpopulation)
3. a live cell with 2 or 3 live neighbours survives
4. a dead cell with exactly 3 live neighbours comes to life

Ending state

- Spatially bounded
  - repeat with cycle 1 (non-empty or empty)
  - repeat with cycle >1
- Spatially unbounded
  - move infinitely
  - grow without limit

Dynamics

tail · loop

> **Important**
>
> - We have no way to predict which outcome a given state will lead to.
> - "It hasn't repeated yet" never tells us it won't.

Examples

- https://academo.org/demos/conways-game-of-life/
- https://playgameoflife.com/lexicon/R-pentomino

Simulation models

- agent-based

Cellular automata work when the basic units react **mechanically** to exogenous conditions. When the units have heterogeneous goals, learn from experience, adapt and react in individual ways, they are referred to as a **complex adaptive system**: the units are **agents**, and their analysis is called *agent-based modelling*.

What ABM relaxes

Traditional economic models assume homogeneity, unbounded rationality and convexity. ABM can drop all three, and represent:

- expectations and learning
- strategic interaction
- adaptation and evolution
- competition for a **common finite resource** (land, quotas, standing timber)

To consider with ABM

- how much information do agents have?
- how exactly do they change behaviour?
- micro-founded, so many more parameters
- stochastic, so many more runs
- **validation becomes hard** (Fagiolo et al. 2007)

There is a strong parallelism with object-oriented programming: agents are objects. The converse does not hold - a land plot is an object in the code but not an agent in the model.

Simulation models

- documenting them: the ODD protocol

The flexibility of simulation models is also their communication problem: with no standard functional form, a reader cannot tell what the model does from an equation.

ODD (Grimm et al. 2020)

The *de facto* standard for **model** documentation:

Overview

1. Purpose and patterns
2. Entities, state and dynamic variables
3. Process overview and scheduling

Design concepts

4. Theoretical background, emergence, adaptation, objectives, learning, sensing, interaction, stochasticity

Details

5. Initialisation
6. Input data
7. Submodels

Some groups use ODD not only to *report* a model but as a **workflow to design** it: writing section 1 honestly, before coding, kills a good number of bad models early.

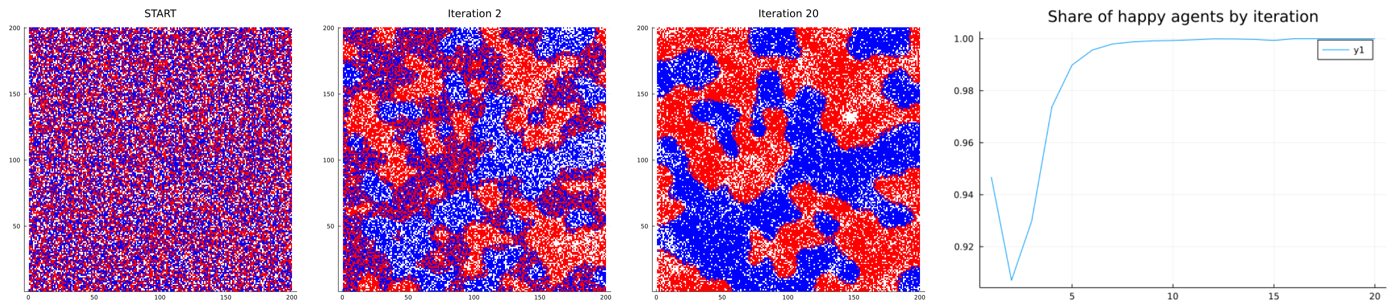
UML — Unified Modeling Language

The *de facto* standard for **implementation** documentation.

- **Class diagram** — the classes, their attributes and methods, and the relations between them (inheritance, composition, association)
- **Sequence diagram** — the flow of messages between objects over time. Useful for documenting the scheduling of a simulation step: who is queried, in which order, and what they answer.
- **Activity / state diagram** — the life cycle of an agent, or the logic of a decision.

Simulation models

- Example 2: the Schelling segregation model



Importance:

- Classical agent-based model in the social sciences
- Introduced the concepts of emerging macro behaviours and tipping points
- One of the very first results obtained from running simulations, especially in the social sciences

How does it work?

- The World is modelled as a gridded space inhabited by two groups (blacks/whites in the original paper)
- Each group has a preference to live in a neighbourhood inhabited by agents of their own type, with a certain "tolerance"
- When the share of agents of its own type in the neighbourhood falls below this "tolerance" level, the agent tries to relocate somewhere with a higher share of its own type of agents.

Results:

- Even a relatively mild preference for the presence of similar agents would have led to a segregated world
- There was a specific threshold level (between 30% and 40%) that was driving these two completely different

Model typologies

- a summary

	Statistical	Predictive / ML	Process-based	Equilibrium	CA / ABM
Primary purpose	explain	predict	explain + project	project policy	explore mechanisms
Mechanism	correlative	correlative	causal	causal (economic)	causal (behavioural)
Extrapolation	poor	very poor	possible	regime-dependent	possible
Heterogeneity	aggregate	any	stand/pixel	representative agent	individual
Uncertainty	built-in (CI)	out-of-sample error	added externally	added externally	added externally
Data need	moderate	high	high (process-level)	moderate	high (micro-level)

None of these is "the best". The question is always: *which one can answer my research question with the data and the time I have?* (Kelly (Letcher) et al. 2013)

Part C

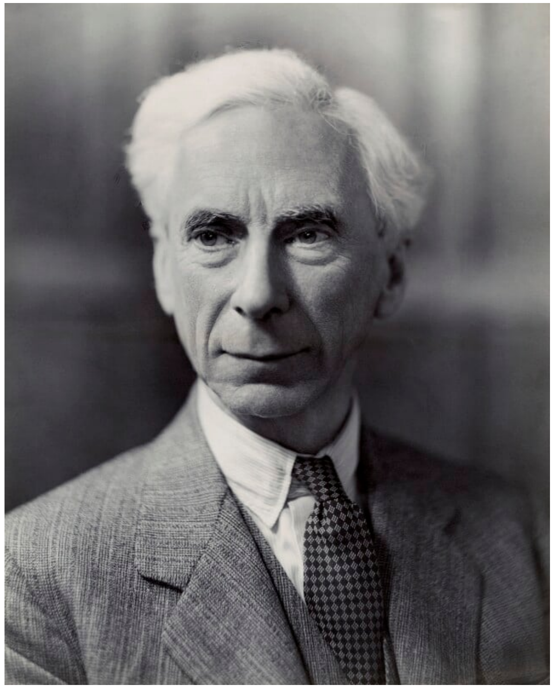
Choosing the right model typology

The Inductivist Turkey

- **Inductivism**: gather facts → make general laws.
- **Deductivism**: start with a theory → test it with facts.

The inductivist turkey (Chalmers 1999; after Russell 1912):

> The turkey found that, on his first morning at the turkey farm, he was fed at 9 a.m. Being a good inductivist turkey he did not jump to conclusions. He waited until he collected a large number of observations that he was fed at 9 a.m. and made these observations under a wide range of circumstances, on Wednesdays, on Thursdays, on cold days, on warm days. Each day he added another observation statement to his list. Finally he was satisfied that he had collected a number of observation statements to inductively infer that "I am always fed at 9 a.m.".
>
> However on the morning of Christmas eve he was not fed but instead had his throat cut.

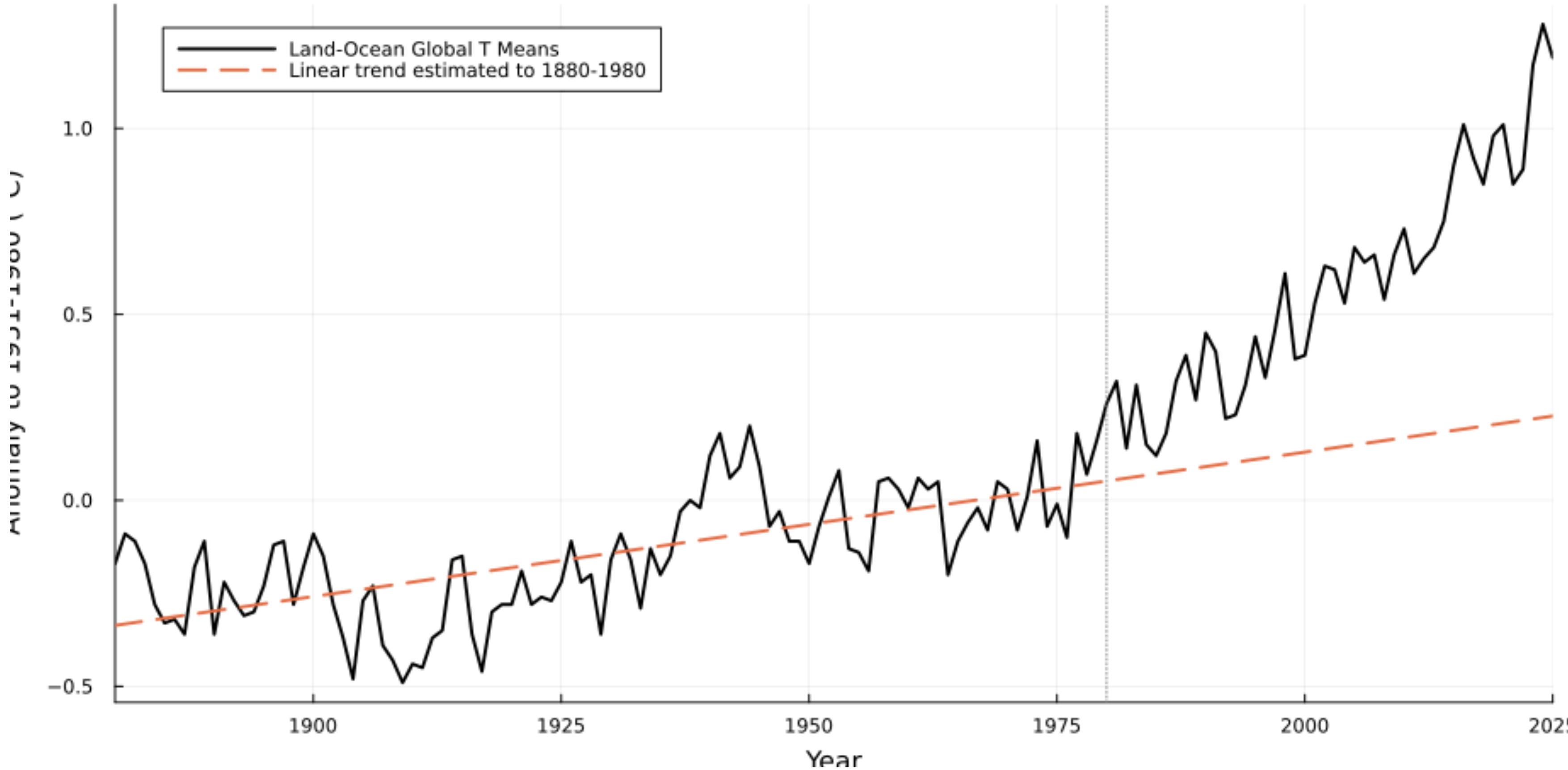


- → look at the **causes** of the trend, not only at the trend itself
- → same argument as the Lucas critique in economics: estimated relations shift when the rules of the game shift, so model the deep parameters instead (Lucas 1976)

Examples: Kuznets curve, Environmental Kuznets curve, peak oil theory (Deming 2023) and the fitted temperature trend of the previous slides

Statistical models

- the extrapolation trap

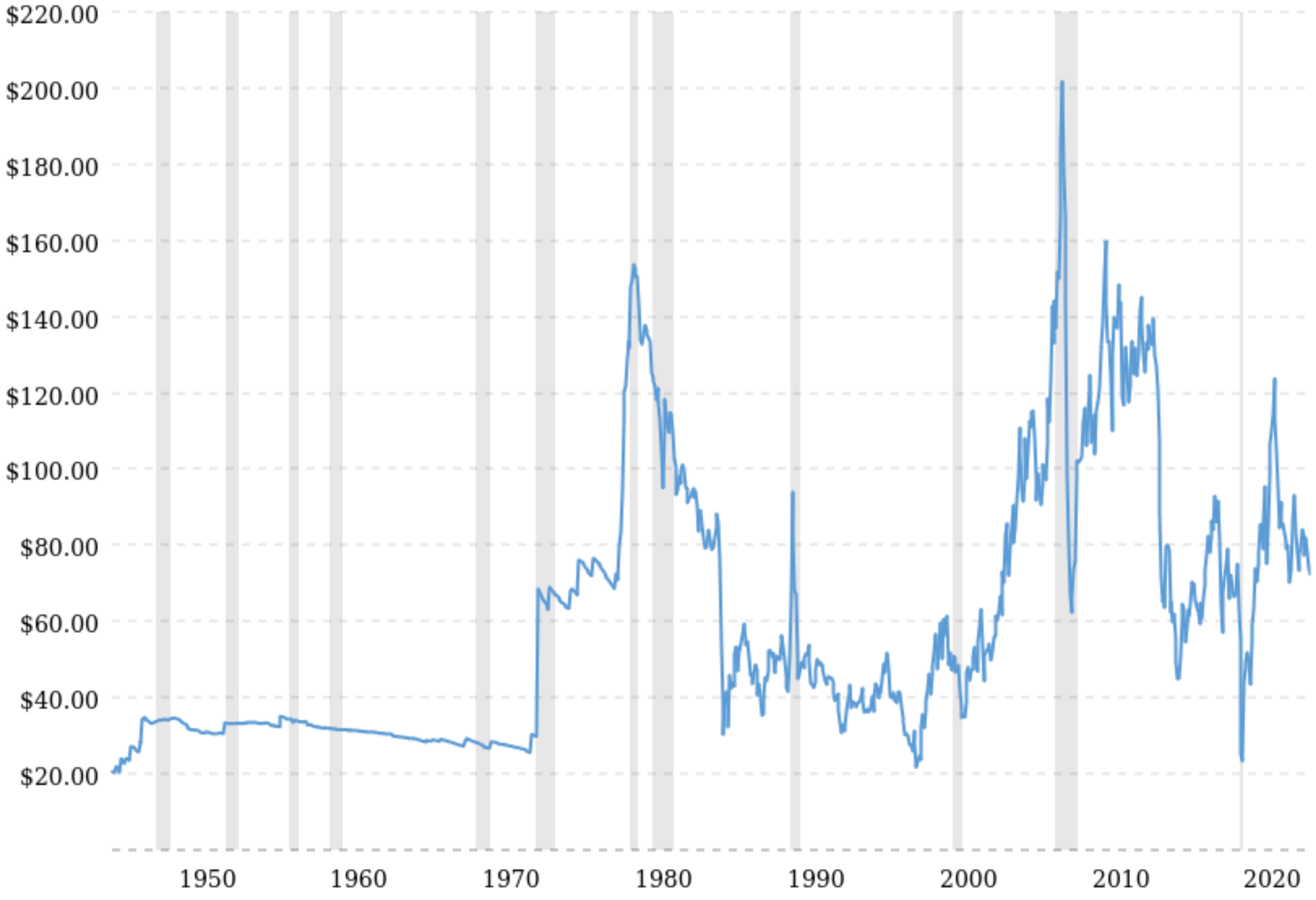


Source: NASA GISTEMP, https://data.giss.nasa.gov/gistemp/

A trend estimated **on a century** of data, extrapolated over the following 45 years, strongly underestimates the anomaly. The fit was excellent; the model was still the wrong tool for the question.

Historical oil prices (and recessions)

West Texas Intermediate (WTI or NYMEX) crude, inflation-adjusted dollars per barrel



Source: https://www.macrotrends.net/1369/crude-oil-price-history-chart

The fear of oil depletion and reaching the "peak oil"

Peak theory: production curves of non-renewing resources approximate a bell curve. Thus, according to this theory, when the peak of production is passed, production rates enter an irreversible decline.

Nixon signs 55 mph U.S. speed limit

THE CINCINNATI ENQUIRER

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Nixon Signs 55 MPH U.S. Speed Limit

(c) The Washington Post

SAN CLEMENTE, Calif. — President Nixon Wednesday signed legislation aimed at forcing states to reduce their speed limits to 55 miles an hour. He claimed the lower limit would save 200,000 barrels of fuel a day.

In another action intended to provde a more long-range energy alternative, the President signed legislation that would restructure the rail system in the Northeast and Midwest and rescue seven failing railroads. The legislation authorizes spending of $1.5 billion in federally guaranteed obligations and $500 million in direct payments that can be used for interim cash payments to bankrupt railroads, for protection of displaced employees and for local rail service subsidies.

The President also signed a $73.7 billion 1974 defense appropriation bill, accepting without comment a cut of $3.5 billion from the amount he originally asked Congress to approve.

The bill, nearly $2 billion less than Congress voted in 1973, provides the funds for military pay, operations, weapons procurement, and research and development in the year ending June 30.

As enacted, few leaders either in government or the military were fully happy with it.

Signing of the measures left Mr. Nixon with only 11 unsigned pieces of legislation still on his desk, including a foreign aid appropriations bill and another bill increasing Social Security payments by 11%. The latter bill is expected to be signed today.

In signing the speed limit bill Mr. Nixon issued a statement in which he praised states for their voluntary efforts and "the response of so many Americans to my request that they slow down on the highways even when the speed limits have not officially been lowered."

The bill will require states, as a condition of continuing to receive their federal shares of highway trust funds, to set a 55 mile an hour speed limit on divided highways of four lanes or more. This also would require states such as Maryland that have set 50 mile an hour limit on these highways to raise them to 55, White House spokesman Gerald L. Warren said in announcing the bill signing.

The maximum limit on other roads under state jurisdiction would be 55.

Another provision of the legislation allows highway funds to be used to pay the cost of carpool demonstration projects. The legislation requires the Department of Transportation to make a report to Congress on methods that would improve the effectiveness of carpooling.

The railroad bill, known as the Regional Rail organization Act of 1973, was hailed by the President as "an important turning point in the history of America's railroad industry."

So Why Work?

WASHINGTON (UPI) — Reorganizing the nation's northeast rail system will wipe out thousands of jobs, but some workers will get as much as $2500 a month for the rest of their lives if they can't find another job.

Under the Railroad Reorganization Act each worker is protected by a unique job severance plan subsidized by the government.

The employee protection plan worked out under the bill would provide monthly checks for employees on a formula based on former earnings, "but not to exceed $2500 per month."

City Loses 102,836 Gallons

Cincinnati's Management Services Office has been unable to find records to account for an additional 102,836 gallons of gasoline bought by

9-Cent Boost In Gas Likely Within Week

From Enquirer Wires

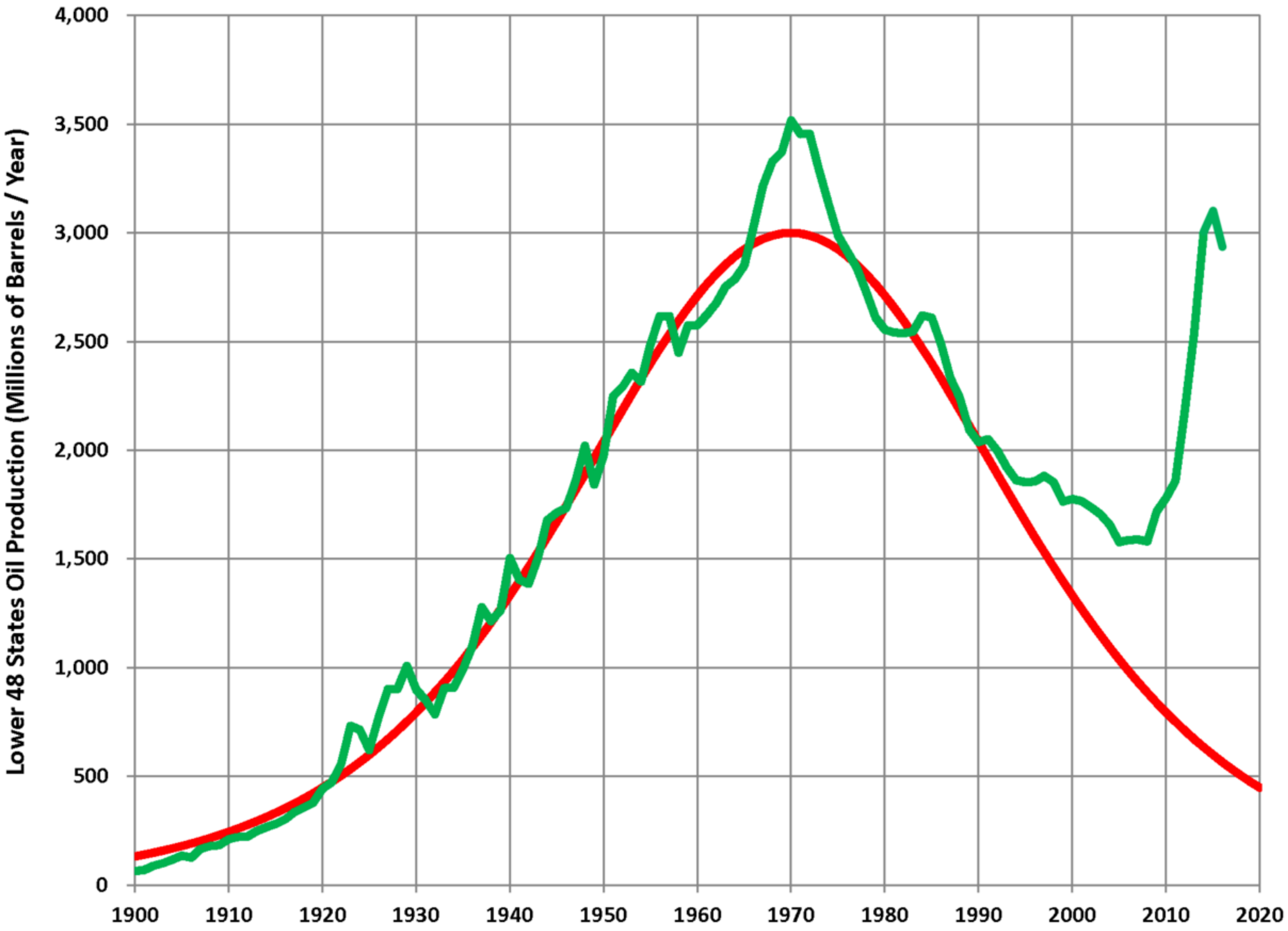
Forecasts of price increases ranging up to nine cents a gallon within a

this fast, but I probably should have," said Binsted. He said the large increases announced by companies such as American reflect skyrocket-

domestic crude oil producers in late December.

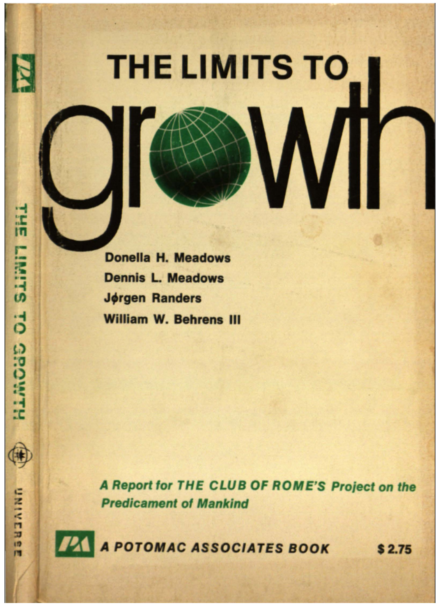
The Greater Cincinnati Gasoline Dealers Association said Wednesday

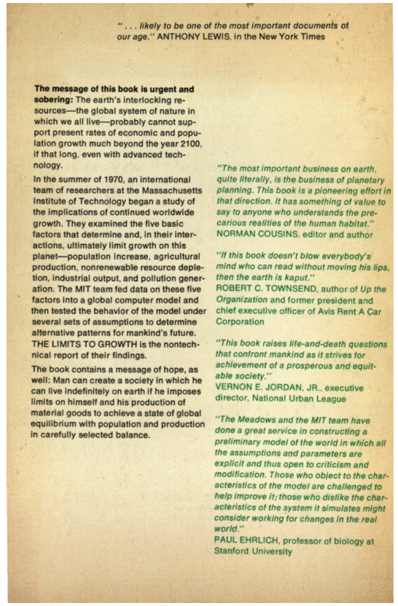
Rise and fall of peak oil theory



Deming (2023), *M. King Hubbert and the rise and fall of peak oil theory*, https://doi.org/10.1306/03202322131

The Limits to Growth Meadows et al. (1972)





"... likely to be one of the most important documents of our age." ANTHONY LEWIS, in the New York Times

The message of this book is urgent and sobering: The earth's interlocking resources—the global system of nature in which we all live—probably cannot support present rates of economic and population growth much beyond the year 2100, if that long, even with advanced technology.

In the summer of 1970, an international team of researchers at the Massachusetts Institute of Technology began a study of the implications of continued worldwide growth. They examined the five basic factors that determine and, in their interactions, ultimately limit growth on this planet—population increase, agricultural production, nonrenewable resource depletion, industrial output, and pollution generation. The MIT team fed data on these five factors into a global computer model and then tested the behavior of the model under several sets of assumptions to determine alternative patterns for mankind's future. THE LIMITS TO GROWTH is the nontechnical report of their findings.

The book contains a message of hope, as well: Man can create a society in which he can live indefinitely on earth if he imposes limits on himself and his production of material goods to achieve a state of global equilibrium with population and production in carefully selected balance.

"The most important business on earth, quite literally, is the business of planetary planning. This book is a pioneering effort in that direction. It has something of value to say to anyone who understands the precarious realities of the human habitat."
NORMAN COUSINS, editor and author

"If this book doesn't blow everybody's mind who can read without moving his lips, then the earth is kaput."
ROBERT C. TOWNSEND, author of *Up the Organization* and former president and chief executive officer of Avis Rent A Car Corporation

"This book raises life-and-death questions that confront mankind as it strives for achievement of a prosperous and equitable society."
VERNON E. JORDAN, JR., executive director, National Urban League

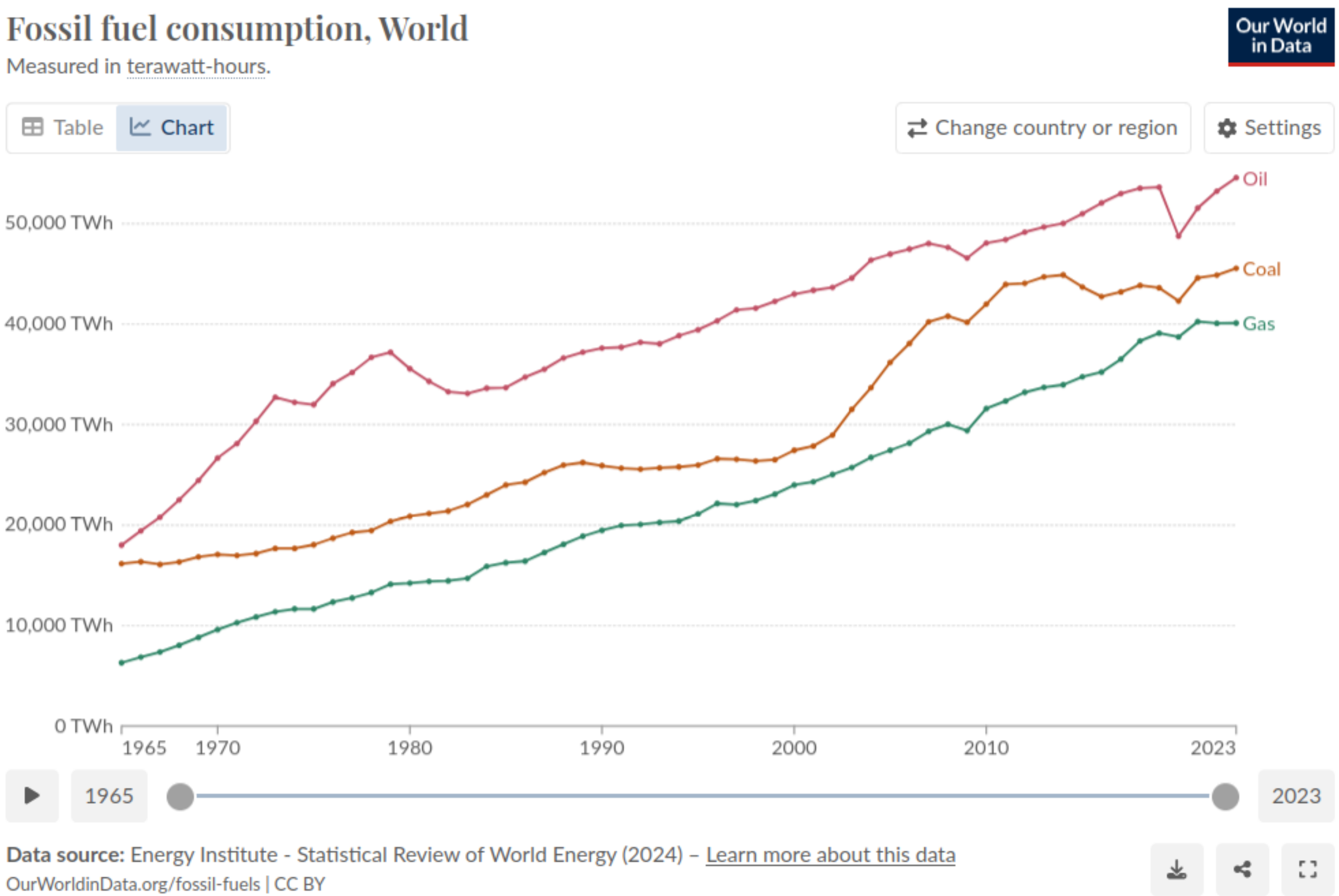
"The Meadows and the MIT team have done a great service in constructing a preliminary model of the world in which all the assumptions and parameters are explicit and thus open to criticism and modification. Those who object to the characteristics of the model are challenged to help improve it; those who dislike the characteristics of the system it simulates might consider working for changes in the real world."
PAUL EHRLICH, professor of biology at Stanford University

> In any finite system there must be constraints that can act to stop exponential growth. These constraints are negative feedback loops. The negative loops become stronger and stronger as growth approaches the ultimate limit, or carrying capacity, of the system's environment. Finally the negative loops balance or dominate the positive ones, and growth comes to an end. In the world system the negative feedback loops involve such processes as pollution of the environment, depletion of nonrenewable resources, and famine.

No learning/adaptation mechanisms!

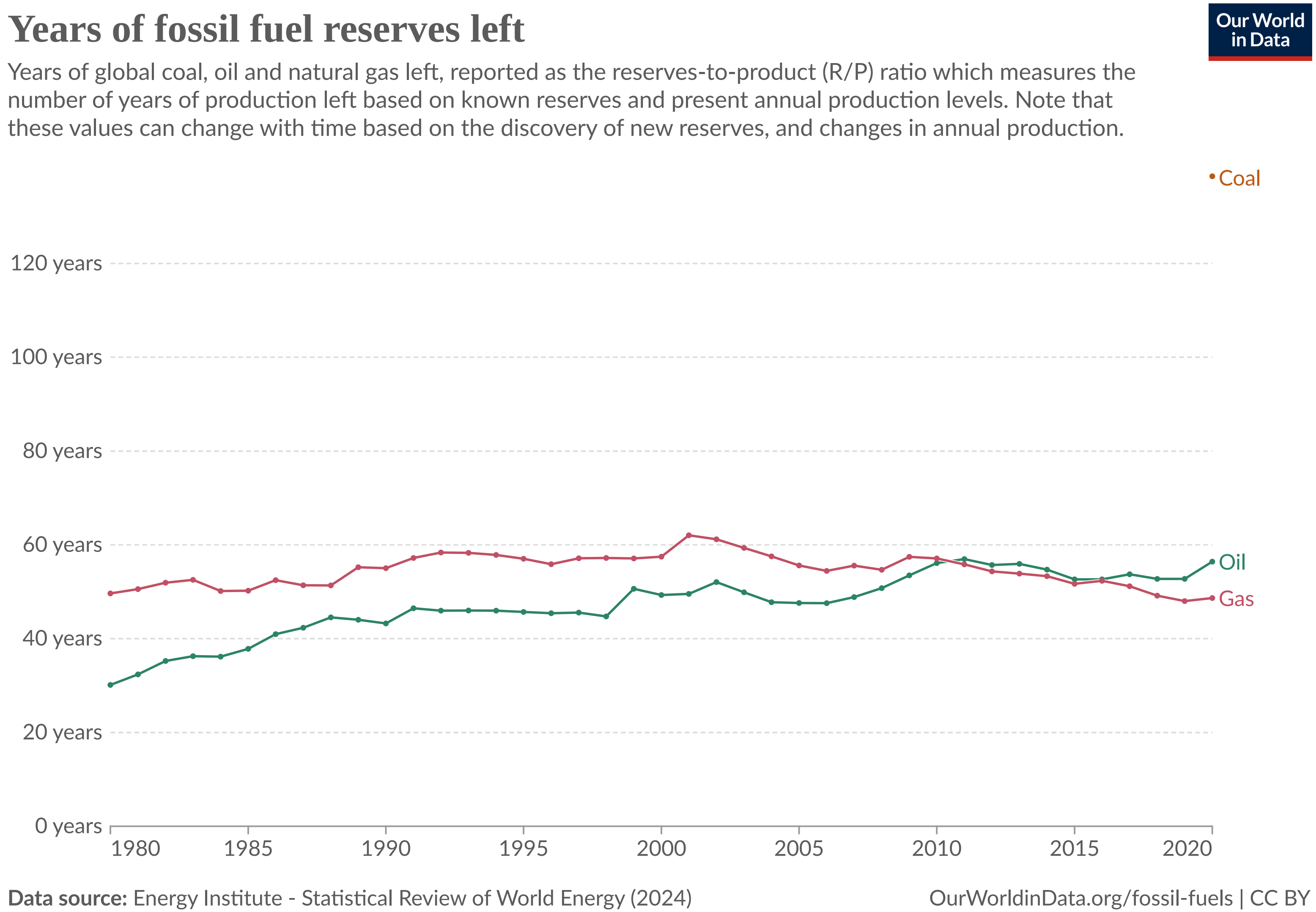
Fossil fuel consumption

- by year and type of fuel



Source: https://ourworldindata.org/fossil-fuels

Fossil fuels reserves



Source: https://ourworldindata.org/grapher/years-of-fossil-fuel-reserves-left

Part D

Model development

The modelling cycle

Model development is **iterative**, and every stage is open to critical review and revision (Jakeman et al. 2006):

1. Define the **purpose** (research question) and the "clients"
2. Specify the **context**: boundaries, scales, resolution
3. Conceptualise the system; choose the model **family**
4. Document the **data**: quantity, quality, limitations
5. Choose the model **structure** and parameters
6. **Estimate** / **calibrate** the parameters
7. **Verify** the implementation (does the code do what the maths says?)
8. **Evaluate** / **validate** against the performance criteria
9. Quantify **uncertainty** and run **sensitivity** analysis
10. Report, and **revise** - go back to step 1

In practice: expect to loop through 3–8 several times, and expect step 1 to change as a result. A model that never changed its research question probably never confronted its data.

Scenarization

A model produces conditional statements. The condition is the **scenario**.

Four terms that are not interchangeable:

- **Forecast, prediction**: the most likely future. Requires a stable regime; rarely defensible beyond a few years.
- **Projection**: a conditional output. *If* the assumptions hold, *then* the model produces this.
- **Scenario**: a coherent, internally consistent story about how the future could unfold, plus the quantification that goes with it.
- **Foresight** (French *prospective*), or forward-looking analysis: the whole exercise of exploring possible futures — building the scenarios, running the models, and working with the people who will use the results.

Börjeson et al. (2006) organise them by the question the *user* asks:

The user asks	**Scenario type**	**Typical technique**
What **will** happen?	**Predictive**	trend extrapolation, econometrics
What **can** happen?	**Explorative**	storyline + simulation
How can a target be reached?	**Normative**	backcasting, optimisation

Scenarization

- in practice

Storyline and simulation: a qualitative narrative, translated into quantitative drivers, fed to the model.

- The reference for the exercise is the SSP-RCP framework (Riahi et al. 2017): five Shared Socioeconomic Pathways (SSP1–SSP5), each crossed with a Representative Concentration Pathway target (the radiative forcing level in W/m^2 reached by 2100)
- For the forest sector, the drivers are usually: GDP and population, energy and carbon prices, climate variables, technology, policy instruments, consumer preferences.

Design rules for a scenario set

- always include a **baseline** (business as usual) - the counterfactual against which everything is read
- vary drivers **jointly and consistently**: high GDP with low energy demand is a story, not a scenario
- span the plausible range rather than clustering around the central case
- keep the set **small enough to communicate**: 4–6 scenarios, named, not numbered
- state explicitly what the model holds **fixed** - that is where the surprises come from

Calibration, verification, validation, evaluation

Four words routinely (and wrongly) used as synonyms (Rykiel Jr. 1996):

- **Calibration**: adjusting parameters so that model output matches observations. It is *fitting*, not testing - you cannot then validate on the same data.
- **Verification**: does the implementation faithfully reproduce the intended mathematics? A **software** question - unit tests, mass/energy balance checks, analytical checks
- **Validation**: a *test*. Does the model meet performance criteria fixed *in advance*, for its *intended use*? Narrow, quantitative, replicable — and not a certificate of truth.
- **Evaluation**: a *judgement*. The whole body of evidence on the model's credibility: are the assumptions defensible, is the structure adequate, do the data support it, does it behave sensibly where no data exist, would a domain expert recognise the system? Broader, partly qualitative, never binary.

Model validation

- what can you test against?

The criterion is only half of a validation design. The other half is **which data the model has never seen**.

- **In-sample fit** – the model reproduces the data it was calibrated on. This is *not* validation, it is a consistency check.
- **Temporal hold-out / hindcasting** – calibrate on 1950–1990, test on 1990–2020. The standard design for climate and forest growth models.
- **Genuine out-of-sample future** – publish the projection, wait, compare. The strongest test, and the one almost nobody runs.
- **Spatial transfer** – parameterise on one region or site network, test on another. Tests whether the parameters mean anything.
- **Independent data source** – inventory plots against remote sensing, declared harvest against mill intake.
- **Cross-validation** – K-fold, or leave-one-site-out when sites are the unit of interest.
- **Limit and analytical cases** – does the model give the known answer where one exists?
- **Cross-model comparison** – model intercomparison exercises, or reproducing another model's published results (*docking*).

Model validation

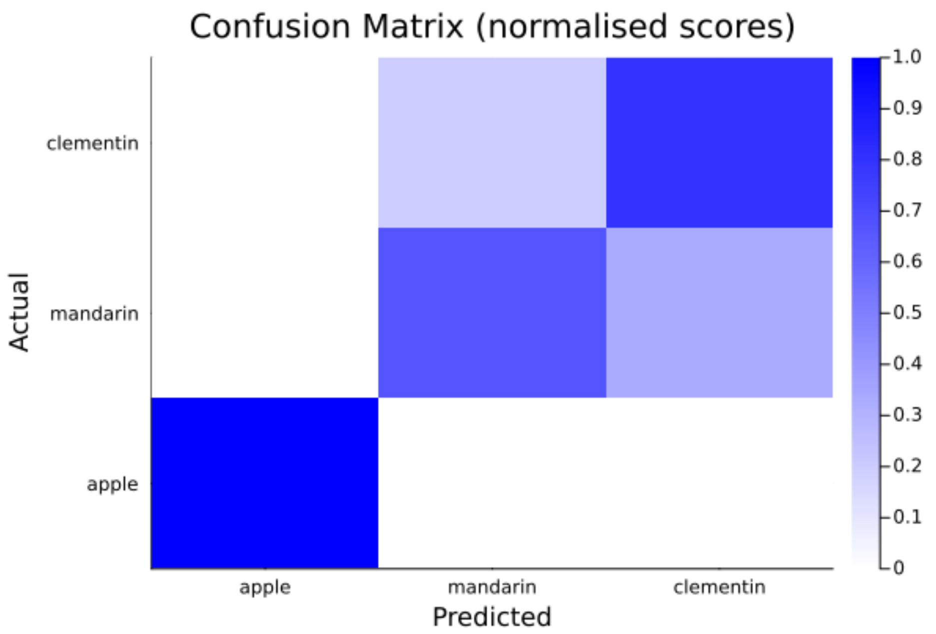
- metrics

Regression / continuous output

- norm-based distances: ℓ_1 (Manhattan), ℓ_2 (Euclidean) - not scaled by sample size, sensitive to units
- **MSE**: ℓ_2 scaled by the number of records
- **Relative mean error**: deviations relative to the mean of the true values - easier to communicate
- **Mean Absolute Scaled Error (MASE)**: the relative mean error of the zero-mean, unit-variance scaled data
- **Sobol' index**: share of the variance of the true data explained by the model; scale-invariant

Classification / categorical output

- **confusion matrix**
- accuracy, precision, recall - which trade off against each other
- with imbalanced classes, accuracy is close to meaningless



Implementations of all of these are in BetaML (Lobianco 2021). We'll go back to them in October.

Model evaluation

- assembling the evidence

Validation is one input. Evaluation is everything you can say about whether the model deserves to be believed (Rykiel Jr. 1996; Jakeman et al. 2006).

On the model itself

- **Assumptions**: are they stated, and are they defensible to someone who knows the system?
- **Structural adequacy**: are the processes that will dominate under the scenarios actually in the model?
- **Behaviour where no data exist**: extreme values, shocks, long runs – does it stay physically and economically sensible?
- **Parameter meaning**: are they measurable independently, or free knobs?
- **Verification**: unit tests, mass and energy balances, analytical cases

On the evidence around it

- **Data audit**: provenance, coverage, measurement error, and what is missing
- **Uncertainty and sensitivity analysis**: reported, and global
- **Expert judgement / face validity**: would a domain expert recognise the system? Structured elicitation, not a corridor conversation
- **Comparison with structurally different models**, not similar ones (Bugmann and Seidl 2022)
- **Track record**: how did earlier applications of this model fare?
- **Transparency**: documentation (ODD), code and data available, limitations stated by the authors themselves

Evaluation is never binary, and it is never finished. It is also the only place where **fitness for the intended decision** can be judged – no metric does that.

Stochasticity, uncertainty and sensitivity analysis

Three different things, routinely conflated.

- **Stochasticity** is *inside* the model: random initialisation, random draws in the processes, random sampling. It generates a distribution of outputs from a single parameter set $\Rightarrow$ you must run **replicates** and report a distribution, never a single run.
- **Uncertainty analysis** asks: *how uncertain is the output?* – assign distributions to the inputs, propagate them through the model, obtain a distribution of the output.
- **Sensitivity analysis** asks: *which inputs drive that uncertainty?* – apportion output variance among input factors.

Refsgaard et al. (2007) classify uncertainty by **nature** and by **location**:

Location	**Example**	**Reducible?**
Input data / parameters	inventory measurement error, mortality rate, price elasticity	partly, with better data
Model structure	which processes are in the model at all	only by building another model
Context / framing	boundaries, scenario assumptions	not by modelling

Sensitivity analysis

- how, and how not

One-at-a-time (OAT)

Move one factor, hold the others at their nominal value. Explores a one-dimensional corridor through a k-dimensional space; blind to interactions. With non-linear models it can rank factors in the wrong order.

Morris screening (Morris 1991)

Randomised one-factor-at-a-time trajectories; cheap; good for *screening* many factors down to a few.

Variance-based (Sobol') (Sobol' 2001)

Decompose output variance into individual inputs and their interactions. Computationally expensive, hence the use of Monte Carlo or quasi-Monte Carlo estimators (and often of emulators/surrogate models).

A systematic review of highly cited papers found that **42%** failed the elementary requirement of exploring the space of input factors at all, moving instead along one-dimensional corridors (Saltelli et al. 2019).

Model integration and coupling

Few interesting questions fit inside one model. Coupling is how the forest sector community answers them - and it is harder than it looks.

Degrees of coupling

- **soft / sequential**: model A writes files, model B reads them. Simple, transparent, no feedback.
- **hard / bidirectional**: A and B exchange state at each time step. Feedbacks captured, complexity explodes.
- **integral** vs **integrated**: one model built as a whole, versus a federation of independent components (Voinov and Shugart 2013)

Where it breaks

- **scale mismatch**: pixel vs region, annual vs decadal
- **semantic mismatch**: the same word meaning different things in two disciplines
- **error propagation**: a local change now propagates through the whole system, defeating the purpose of modularity
- **validation**: each component may be validated; the assembly usually is not

- External integration (files): easy but computationally expensive
- Internal integration (Application Programming Interfaces): harder to implement, but computationally cheaper

> Treating models only as software gives birth to **"integronsters"**: constructs that are perfectly valid as software products but useless as models (Voinov and Shugart 2013).

Reproducibility

Reproducibility is the **minimum standard** for judging a computational claim when full independent replication is impossible (Peng 2011). A published figure without the code and data that produced it is an assertion.

The spectrum

publication only → + code → + data → + linked and executable code and data → full replication

Most modelling papers still sit at the left end.

Practical rules (Sandve et al. 2013)

- record **how** every result was produced
- avoid manual data manipulation: automate data retrieval
- archive the **exact versions** of all external programs/libraries (Project.toml vs Manifest.toml)
- version-control all custom scripts
- record **random seeds**
- store raw data behind every plot

```julia
using Random
Random.seed!(123); println(rand(3))
Random.seed!(124); println(rand(3))
Random.seed!(123); println(rand(3))
```

```
[0.521213795535383, 0.5868067574533484, 0.8908786980927811]
[0.865674388682704, 0.5593067117763364, 0.7174253143031734]
[0.521213795535383, 0.5868067574533484, 0.8908786980927811]
```

⇒ This is not bureaucracy: it is the reason the next two blocks of the course exist (git, environments, testing, literate programming, packaging).

Take-home messages

1. **Start from the research question**, not from the model you know how to build. "All models are wrong, but some are useful" (Box 1976) - useful *for something*, which you must name.
2. **A model type is a set of design choices.** Describe the rows of the grid, not the label on the box.
3. **Correlative models interpolate; causal models extrapolate** - at the price of more assumptions, more parameters and more data.
4. **Fit is not validity.** Define your performance criteria before you test, relative to the intended use (Rykiel Jr. 1996); for open systems, confirmation is always partial (Oreskes et al. 1994).
5. **A single run is not a result.** Report distributions, run a *global* sensitivity analysis, and be explicit about structural uncertainty.
6. **Coupling multiplies the problems** as well as the scope. Beware integronsters.
7. **If it is not reproducible, it is not a scientific result.**

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